

Replication Debate Classroom Run Kit

TOK Cognitive Science Activity Bank • Human Sciences • 35-45 min

Category	Human Sciences	Time	35-45 min
Difficulty	Extended	ID	replication-debate

Big idea: Human science knowledge is context-sensitive.

One-Minute Teacher Brief

Students evaluate what it means for a human-science claim to be supported.

Student Prompt

Inspect Replication Debate Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Replication Debate Stimulus A	Student-facing stimulus 1: A headline claim, a replication attempt, and a contested interpretation. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Replication Debate Stimulus B	Student-facing stimulus 2: Role cards for original researcher, replicator, journalist, and funding body. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.	Use this for comparison, a second round, or a partner challenge.
Replication Debate Reveal / Twist	Student-facing stimulus 3: A reliability scale separating failure, boundary condition, and fraud. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.	Teacher reveal: connect the changed response to the big idea — Human science knowledge is context-sensitive.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Replication Debate Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Replication Debate Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Does failed replication destroy knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Human Sciences.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how human science knowledge is context-sensitive.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Does failed replication destroy knowledge?

Setup Checklist

- Prepare a starter stimulus using: A headline claim, a replication attempt, and a contested interpretation.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- A headline claim, a replication attempt, and a contested interpretation.
- Role cards for original researcher, replicator, journalist, and funding body.
- A reliability scale separating failure, boundary condition, and fraud.

Resources To Use

- Resource pack: <resources/replication-debate/index.html>
- Card deck CSV: <resources/replication-debate/cards.csv>
- Visual prompt: <resources/replication-debate/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Give students a short description of the claim.
2. Groups design a replication study.
3. They must state sample, procedure, outcome measure, possible confounds and what would count as a failure.
4. Debate whether one failed replication should change belief.

Debrief Moves

- Knowledge question: Does failed replication destroy knowledge?
- Why are humans difficult to study?
- What makes evidence strong in psychology or economics?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...